

Aaron Legg

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Education

Cornell University, College of Engineering

B.S. Chemical Engineering and Operations Research

Ithaca, NY

Aug 2024 – May 2028

Scottsdale Community College

A.S. General Engineering, Mathematics, General Studies

Scottsdale, AZ

Aug 2023 – May 2024

Relevant Coursework: Probability & Statistics I & II, Data Structures, Stochastic Calculus & Processes, Foundations of AI Decisions

Projects

HMM Regime Trading Bot ↗

Jan 2025

- Built a modular, vectorized Python backtesting engine with Gaussian-HMM forward-algorithm filtered inference and BIC model selection, eliminating look-ahead bias across 10 equities
- Capped max drawdown under 10% via a volatility-rank allocation algorithm (60–95% exposure, 1.25x leverage) across 10 equities
- Shipped a separate risk layer with tiered circuit breakers, 60-day correlation caps, and 100-run Monte Carlo cash-gap stress testing

Monte Carlo Option Pricer ↗

Jan 2025

- Built a high-performance European option pricing engine in Java on geometric Brownian motion, hitting 0.01% error versus the Black-Scholes closed form over 1M+ stochastic paths
- Improved computation speed 35% by optimizing Box-Muller sampling to O(N) time and O(1) space; validated via put-call parity and EFX futures comparison

Voice Controlled Ledger

Mar 2026

- Architected an event-driven 3-model AI pipeline (Whisper, GPT, Claude) on containerized FastAPI at sub-3s speech-to-ledger latency
- Cut manual ledger-entry time 80% with an asynchronous NLP interface parsing amount and FX across 99 languages
- Integrated 5 Google APIs (OAuth 2.0, Sheets, Docs, Calendar) behind a unified, fault-tolerant token-refresh service with least-privilege scoping

Experience

Cornell Trading

Ithaca, NY

Quantitative Chair

Aug 2026 – Present

- Designed and taught an 8-week quant curriculum on Python, statistical arbitrage, and backtesting to 43 new members
- Directed 6 research teams and standardized a shared Python backtesting framework, cutting strategy-development time 30%
- Mentored members through a paper-trading competition; top teams reached a 1.5 Sharpe over 10 weeks

Fry's Food Stores

Scottsdale, AZ

Operations Intern – Supply Chain Management

June 2026 – Aug 2026

- Raised ordering efficiency 18% by deploying an AI agent that autonomously reordered out-of-stock products from POS data
- Cut void transactions per shift 40% by training 20+ associates on tender, coupon, and price-check standards
- Drove till variance under \$2/shift by auditing 5+ tills daily across 3 registers

Tiny Smiles

Scottsdale, AZ

Risk Management Intern

May 2025 – Aug 2025

- Ran vectorized Monte Carlo risk simulations on six-figure investment accounts, quantifying VaR and drawdown distributions
- Reduced downside-risk exposure 18% through performance attribution and asset-allocation optimization in Excel/VBA

Teaching Experience

Cornell University – SYSEN 6170

Ithaca, NY

Teaching Assistant – Quantitative Foundations of Systems Engineering

Aug 2026 – Present

- Led office hours and recitations for 30 PhD students, resolving 100+ convex-optimization questions
- Graded problem sets and modeling case studies on Python/MATLAB optimization formulations; mentored 8 teams through capstone model formulation, validation, and policy analysis

Cornell University – CEE 5980

Ithaca, NY

Teaching Assistant – Decision Framing and Analytics

Aug 2025 – Present

- Supported 30+ graduate students in office hours and graded homework and quizzes on decision trees and multi-objective analytics
- Advised teams on decision-analysis capstones spanning problem framing, alternative generation, and triple-bottom-line evaluation

Technical Skills

Languages & Tools: Python, C/C++, Java, Bash, SQL, Git, Perforce, Linux, Docker, ReactJS

Libraries: NumPy, Pandas, scikit-learn, statsmodels, PyTorch, Hugging Face Transformers, FastAPI

Quant/ML: Monte Carlo methods, Hidden Markov Models, backtesting, time-series modeling, supervised learning